23A01601 · DESIGN OF STEEL STRUCTURES

UNIT – I

Introduction to Structural Steel and Design of Connections

Syllabus: General – Types of Steel – Properties of Structural Steel – I.S. Rolled Sections – Concept of Limit State Design – Design of Simple and Eccentric Bolted and Welded Connections – Types of Failure and Efficiency of Joint – Prying Action – Introduction to HSFG bolts.

What's inside these notes

- Types of steel & properties (IS 2062 grades)
- IS rolled sections & section classification
- Limit State Design – safety factors, load combinations
- Bolted connections: bearing bolts, capacities, failures, efficiency
- Eccentric bolted connections (torsion + shear, bending + shear)
- Welded connections: fillet & butt, simple and eccentric
- Prying action · HSFG bolts · solved examples

Exam tip: Every connection problem = (1) find capacity of ONE bolt/weld, (2) divide load, (3) check plate. Write γ_m values clearly – marks are given for them.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2\times10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Introduction to Structural Steel

Steel = alloy of **iron + carbon (0.12 – 0.25 % for structural steel)** + small amounts of Mn, Si, S, P. Used for buildings, bridges, towers, industrial sheds, transmission lines, silos, chimneys etc.

Advantages

- **High strength-to-weight ratio** → light structures, smaller foundations
- **Ductile** – large deformation before failure (gives warning)
- Uniform quality, **fast erection**, prefabrication in shop
- Can be **reused / recycled**; easy to strengthen or modify later
- Long spans and slender members possible; good toughness & fatigue strength

Disadvantages

- Corrosion → painting / galvanising needed (maintenance cost)
- Loses strength at high temperature (≈ 50 % at 550 °C) → fire protection
- Slender members buckle → bracing needed; fatigue & brittle fracture at low temp.

Types of structural steel (IS 2062)

Type	Grade / designation	f_y (MPa)	f_u (MPa)
Mild steel	E250 (Fe 410)	250	410
High-tensile (HSLA)	E300 (Fe 440)	300	440
HSLA	E350 (Fe 490)	350	490
HSLA	E410 (Fe 540)	410	540
Very high strength	E450 (Fe 570)	450	570

> Quality grades A, B, C (BR, B0): differ in **killed/semi-killed**, impact test & weldability. Grade A – no impact test; B – 25 °C test; C – 0 °C. In E250 (Fe 410W) "W" = weldable.

Other types: **cast steel** (complex shapes, bearings), **stainless steel** (corrosion resistant), **weathering steel / COR-TEN** (forms protective rust layer), **cold-formed steel** (thin sections, IS 801), **wrought iron/cast iron** (obsolete).

2. Properties of Structural Steel

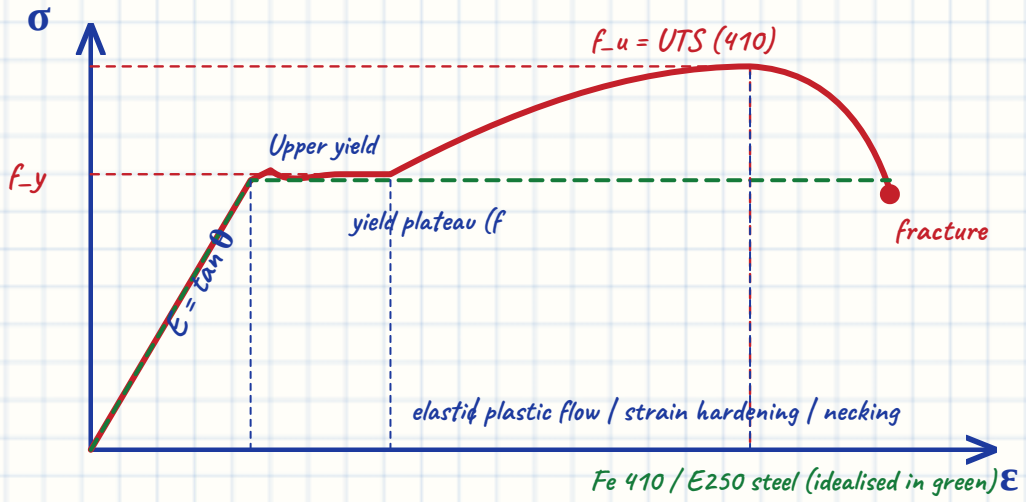
Physical (IS 800 – Cl. 2.2.4)

Property	Value
Unit weight (density)	7850 kg/m³ (78.5 kN/m³)
Modulus of elasticity E	2×10^5 MPa (2.0×10^5 N/mm²)
Shear modulus G	0.769×10^5 MPa
Poisson's ratio μ	0.3 (elastic range)
Coeff. of thermal expansion α	12×10^{-6} per °C

$G = E / [2(1+\mu)]$

Mechanical

- **Yield stress f_y** – design stress in tension/compression
- **Ultimate (tensile) strength f_u** – used for net-section & bolt checks
- **Ductility** – % elongation ≥ 23 % (gauge length $5.65\sqrt{A_0}$) for Fe 410
- **Toughness** – energy absorbed in Charpy test; resists brittle fracture
- **Hardness, weldability, fatigue strength**, residual stress from rolling/welding



Stress–strain curve of mild steel

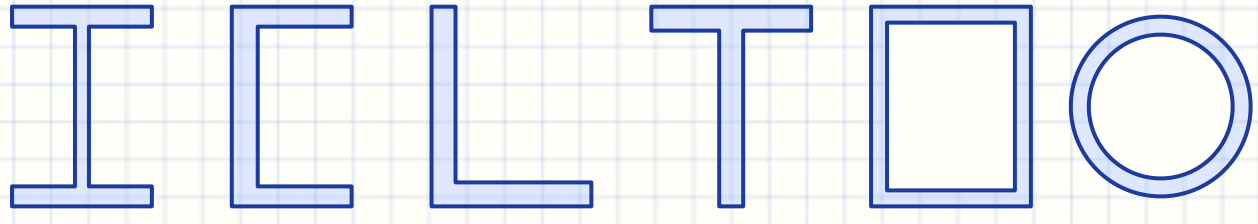
> Mild steel shows a **sharp yield point + yield plateau** ($\varepsilon \approx 0.2\ \% \rightarrow 2\ \%$), strain hardening up to f_u, then necking. High-tensile steel has **no clear yield point** → yield taken as **0.2 % proof stress**.

Why ductility matters

Ductile steel redistributes stress at holes/connections and allows **plastic analysis**; it warns before collapse.

3. I.S. Rolled Steel Sections

Hot-rolled sections are standardised (IS 808, properties in **SP-6 (1)** steel tables) and available in lengths up to 12–15 m.



Common rolled and hollow sections

Section	Designation example	Use
I-beams	ISMB 300 (medium weight), ISLB, ISJB (junior), ISWB (wide)	beams, columns
Heavy beam / column	ISHB 300	columns (bf ≈ depth)
Channels	ISMC 150, ISLC	purlins, built-up members
Angles	ISA 90×90×8 (equal), ISA 100×75×8 (unequal)	truss members, bracing
Tee	ISNT, ISST	chords of light trusses
Flats / plates	ISF 100×10	gussets, cover plates
Hollow	Tube (CHS), SHS, RHS (IS 4923)	space frames, columns

Designation: **ISMB 300 @ 44.2 kg/m** → I-section, Medium-weight Beam, depth 300 mm, mass 44.2 kg/m.

Section properties to read from tables

A (area), I_{zz}, I_{yy}, r_{zz}, r_{yy}, Z_{ez}, Z_{pz}, t_f, t_w, root radius R_1.

Example ISMB 300: A = 5626 mm², I_{zz} = 8603.6 cm⁴, I_{yy} = 453.9 cm⁴, Z_{ez} = 573.6 cm³, r_{yy} = 28.4 mm.

Section classification (IS 800 Table 2) – $\varepsilon = \sqrt{(250/f_y)}$

Class	Behaviour	Moment capacity
1 Plastic	forms plastic hinge + rotation capacity	$Z_p f_y/\gamma_{m0}$
2 Compact	reaches M_p, limited rotation	$Z_p f_y/\gamma_{m0}$
3 Semi-compact	reaches yield in extreme fibre only	$Z_e f_y/\gamma_{m0}$
4 Slender	local buckling before yield	effective section

Limits (rolled I): flange outstand $b/t_f \le 9.4\varepsilon$ (plastic)… 10.5ε (compact)… 15.7ε (semi-compact); web $d/t_w \le 84\varepsilon$ / 105ε / 126ε.

4. Concept of Limit State Design (LSM)

Limit state = condition beyond which the structure no longer satisfies the design requirement. IS 800:2007 is based on LSM (earlier IS 800:1984 used Working Stress).

Two groups

- **Limit State of Strength (collapse)** – yielding, rupture, buckling, fracture, fatigue, overturning, sway stability → use **factored loads**
- **Limit State of Serviceability** – deflection, vibration, durability, fire resistance, corrosion → use **unfactored (service) loads**

Design format

$$\text{Design action} \le \text{Design strength}$$

$$S_d \le R_d = R_u / \gamma_m$$

Partial safety factors apply on **loads (γ_f)** and on **material strength (γ_m)**.

Partial safety factors for materials (IS 800 Table 5)

Limit state	γ_m
Yielding, buckling (γ_{m0})	**1.10**
Ultimate stress / rupture (γ_{m1})	**1.25**
Bolts – bearing type (γ_{mb})	1.25
HSFG bolt – slip (γ_{mf})	1.25 (1.10 if slip is a serviceability limit)
Shop welds (γ_{mw})	**1.25**
Field welds (γ_{mw})	**1.50**

Load combinations for strength (IS 800 Table 4)

$$DL + LL \rightarrow 1.5\,DL + 1.5\,LL$$

$$DL + LL + WL/EL \rightarrow 1.2\,(DL + LL + WL)$$

$$DL + WL/EL \rightarrow 1.5\,DL + 1.5\,WL \text{ (and } 0.9\,DL + 1.5\,WL \text{ for uplift/stability)}$$

Accidental / crane loads – see code

Deflection limits (IS 800 Table 6, short list)

Member	Limit
Floor beams (supporting plaster, LL)	span/360
Other floor / roof members	span/300
Purlins, girts – elastic cladding	span/150
Purlins, girts – brittle cladding	span/180
Gantry girder – EOT crane (< 50 t)	span/750
Gantry girder – manual crane	span/500

Limit-state advantage: same reliability for all members, economical, treats loads and materials separately.

5. Bolted Connections – Basics

Connections join members and transfer force. **Bolts, rivets (obsolete), welds.**

Types of bolts

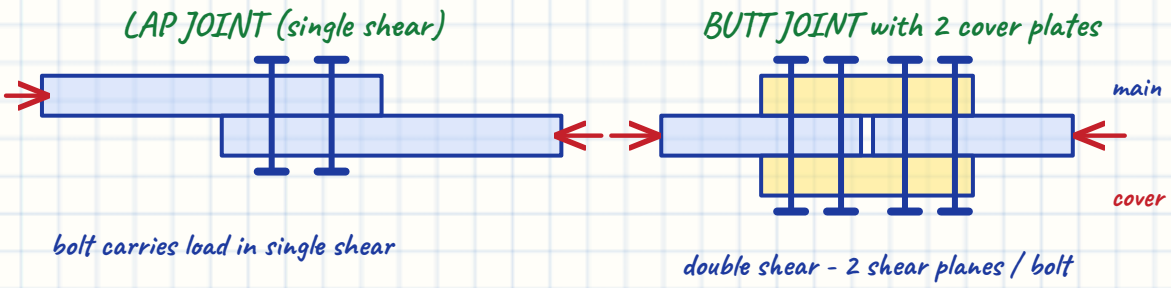
- **Black (unfinished) bolts** – grade 4.6, hole clearance, low cost – light structures, secondary members
- **High-strength bolts, bearing type** – grade 8.8 / 10.9 – shear carried by bolt in bearing
- **HSFG bolts** – tightened to give pre-tension; load transferred by **friction** (no slip)

Grade "4.6": $f_{ub} = 4 \times 100 = 400$ MPa, $f_{yb} = 0.6 \times 400 = 240$ MPa. Grade 8.8: $f_{ub} = 800$, $f_{yb} = 640$. Grade 10.9: $f_{ub} = 1000$, $f_{yb} = 900$.

Joint types

- **Lap joint** – plates overlap → single shear + eccentricity (bending)

- **Butt joint** – plates butt, cover plate(s) on one or both sides → single / **double shear**



Lap joint and Butt joint

Bolt hole diameter d_0 (IS 800 Table 19)

Bolt dia d	d_0
12 – 14 mm	d + 1 mm
16 – 22 mm	d + 2 mm
≥ 24 mm	d + 3 mm

Pitch and edge distance (IS 800 Cl. 10.2)

- **Min. pitch / gauge** $p, g \geq 2.5\ d$
- **Max. pitch** ≤ min(32 t, 300 mm) tension / compression-outer rows: 16 t or 200 mm tension; 12 t or 200 mm compression (t = thinner plate)
- **Min. edge/end distance** $e \geq 1.7\ d_0$ (rolled, machine flame-cut) ; $1.5\ d_0$ (sheared / hand flame-cut)
- **Max. edge distance** $\leq 12\ t\ \varepsilon$ ($\varepsilon = \sqrt{(250/f_y)}$)

> Why these limits? Small pitch → bearing/tear-out failure & hard to tighten; big pitch → plates gape, corrosion, buckling of compression plates.

6. Strength of a Bearing Bolt (IS 800 Cl. 10.3)

(a) Shear capacity

$$V_{dsb} = V_{nsb} / \gamma_{mb} ; V_{nsb} = (f_{ub}/\sqrt{3})\ (n_n A_{nb} + n_s A_{sb})$$

n_n = no. of shear planes **through threads**, n_s = through shank. A_{nb} = net tensile area at threads ≈ $0.78 \times \pi\ d^2/4$, A_{sb} = shank area $\pi\ d^2/4$.

Take $f_{ub}/\sqrt{3}$ and divide by $\gamma_{mb} = 1.25$.

(b) Bearing capacity (on plate)

$$V_{dpb} = V_{npb} / \gamma_{mb} ; V_{npb} = 2.5\ k_b\ d\ t\ f_u$$
$$k_b = \text{smaller of } e/(3d_0)\ ,\ p/(3d_0) - 0.25\ ,\ f_{ub}/f_u\ ,\ 1.0$$

(t = sum of thicknesses of plates bearing in the same direction; use f_u **of plate**.)

(c) Bolt value = min(shear, bearing)

Number of bolts $n = P_u$ / bolt value.

(d) Tension capacity of bolt

$$T_{db} = T_{nb}/\gamma_{mb} \; ; \; T_{nb} = 0.9\, f_{ub}\, A_{nb} \le f_{yb}\, A_{sb}\, (\gamma_{mb}/\gamma_{m0})$$

(e) Combined shear + tension

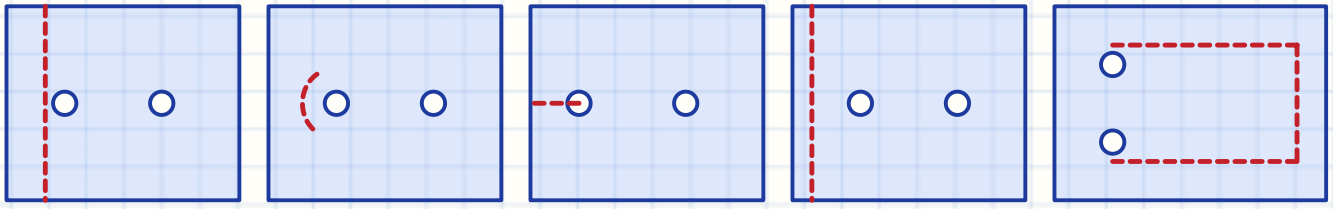
$$(V_{sb}/V_{db})^2 + (T_b/T_{db})^2 \le 1.0$$

Reduction factors

- **Long joint** ($l_j > 15\,d$): $\beta_{lj} = 1.075 - l_j/(200\,d)$ $(0.75 \le \beta_{lj} \le 1.0)$ – bolts at ends carry more
- **Large grip** ($l_g > 5\,d$): $\beta_{lg} = 8d/(3d + l_g)$
- **Packing plates** $t_{pk} > 6$ mm: $\beta_{pk} = 1 - 0.0125\, t_{pk}$

> Bolts in **double shear** have two shear planes → capacity ≈ 2× (but check bearing on the thinner plate set).

7. Types of Failure of a Bolted Joint



Failure modes of a bolted lap joint

Failure	Cause	Prevented by
1. Shear of bolt	bolt too small / few	more / bigger bolts, double shear
2. Bearing of plate (or bolt)	thin plate, small edge dist.	thicker plate, bigger d
3. Edge tear-out (shear-out)	edge/end distance too small	$e \ge 1.5$–$1.7\, d_0$
4. Tension failure of plate at net section	holes reduce area	adequate width, staggered holes
5. Block shear	tearing out a block of plate along bolt line	check T_{db} (see Unit II)
6. Bolt tension / prying	tension joints	thicker flange, more bolts
7. Buckling of plate	compression, large pitch	pitch limits

Net area of a plate

$$A_n = (b - n\,d_0)\,t \text{ (n holes in the section)}$$

For staggered holes: > $A_n = [\,b - n\,d_0 + \Sigma\, p^2/(4g)\,]\,t$

(p = stagger pitch, g = gauge between the two holes.)

Strength of plate

$$\text{Yielding of gross section: } T_{dg} = A_g\, f_y / \gamma_{m0}$$
$$\text{Rupture of net section: } T_{dn} = 0.9\, A_n\, f_u / \gamma_{m1}$$

Efficiency of a joint

$$\eta = \text{Strength of joint / Strength of solid plate} = (\text{min of } P_{ds}, P_{db}, P_{dn}) / (A_g\, f_y/\gamma_{m0})$$

Always < 100 %. Typical 60 – 85 %. Improve with double cover plate (double shear), staggered/zig-zag bolts.

8. Design of Simple Bolted Connection – Examples

EXAMPLE 1 – STRENGTH + EFFICIENCY OF A LAP JOINT

Two plates 100 × 10 mm (Fe 410) lapped, joined by **3 bolts M16, grade 4.6**, in one line along the load. e = 30 mm, p = 40 mm. Find joint strength & efficiency.

Bolt: d_0 = 18 mm, A_{sb} = 201 mm², A_{nb} = 0.78 × 201 = 157 mm².

Shear (single, thread in plane): V_{dsb} = (400/√3) × 157 / 1.25 = **29.0 kN**

Bearing: k_b = min(30/54 = 0.556, 40/54 − 0.25 = 0.49, 400/410 = 0.976, 1) = **0.49**

V_{dpb} = 2.5 × 0.49 × 16 × 10 × 410 /1.25 = 64.3 kN

Bolt value = 29.0 kN → 3 bolts : P_{ds} = **87.0 kN**

Plate: T_{dg} = 1000 × 250/1.1 = 227.3 kN ; A_n = (100−18)×10 = 820 mm² → T_{dn} = 0.9×820×410/1.25 = 242.1 kN

Strength of joint = **87.0 kN** (bolt shear governs)

$$\text{Efficiency } \eta = 87.0 / 227.3 = 0.38 = 38\ \%$$

EXAMPLE 2 – NUMBER OF BOLTS FOR A BUTT JOINT

Butt joint of 2 plates 200 × 12 mm with double cover plates 2 × 8 mm, M20 (4.6) bolts, to transmit **P_u = 300 kN**.

A_{sb} = 314, A_{nb} = 245 mm². **Double shear:** V_{dsb} = 2 × (400/√3) × 245 / 1.25 = **90.6 kN**

Bearing on main plate t = 12 (vs. 2 × 8 = 16 for covers → use 12): d_0 = 22 ; take e = 40, p = 60 : k_b = min(0.606, 0.659, 0.976, 1) = 0.606.

V_{dpb} = 2.5 × 0.606 × 20 × 12 × 410 / 1.25 = **119.3 kN**

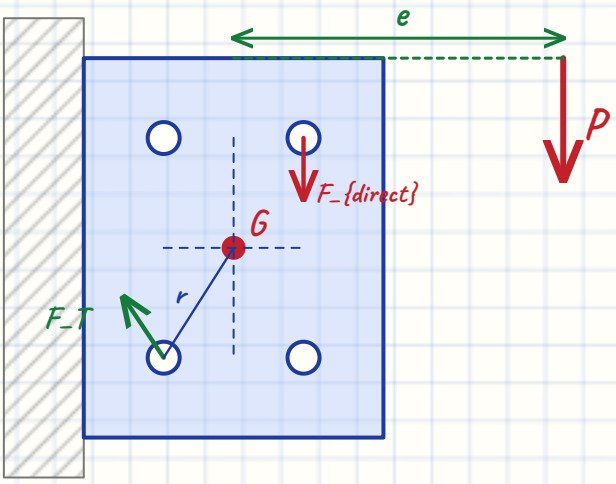
Bolt value = 90.6 kN → n = 300 / 90.6 = 3.3 → provide **4 bolts** on each side of the joint.

Check: $p \geq 2.5d = 50$ ✓, $e \geq 1.7\, d_0 = 37.4$ ✓.

9. Eccentric Bolted Connections

When the line of action of load does **not pass through the centroid of the bolt group** → extra moment $M = P \cdot e$.

Case A – Load in the plane of the bolt group (TORSION + direct shear)



Torsion $T = P \cdot e$

$F_T = T\, r / \Sigma r^2$

$F_d = P / n$

resultant = vector sum

Bracket connection – bolt group loaded eccentrically

Steps:

- Locate centroid G of bolt group.
- Direct shear on each bolt: $F_d = P/n$ (parallel to P).
- Torsion $T = P \cdot e$. Force on bolt i: $F_{T,i} = T\, r_i / \Sigma\, r_i^2$ ($\perp$ to r_i).
- Add vectors – resultant on the **worst bolt** $R = \sqrt{(F_x^2 + F_y^2)}$. Must be ≤ bolt value.

$$F_{T,i} = (P\, e)\, r_i / \Sigma (x^2 + y^2)$$

Components: horizontal = $F_T \cdot (y/r)$, vertical = $F_T \cdot (x/r)$.

EXAMPLE 3 – BRACKET WITH 4 BOLTS

Factored load P = 60 kN at e = 150 mm. 4 bolts at (±50, ±40) mm from G. Plate 10 mm, Fe 410, M22 (4.6) bolts.

$n = 4 \rightarrow F_d = 60/4 = 15$ kN (↓). $T = 60 \times 150 = 9000$ kN·mm

$r = \sqrt{(50^2 + 40^2)} = 64.03$ mm ; $\Sigma r^2 = 4 \times 4100 = 16400$ mm²

$F_T = 9 \times 10^6 \times 64.03 / 16400 =$ **35.1 kN**

Vertical comp. = 35.1 × 50/64.03 = 27.4 kN ; Horizontal = 35.1 × 40/64.03 = 21.95 kN

Critical bolt (both vertical parts add): V = 15 + 27.4 = 42.4 kN, H = 21.95 kN

$$R = \sqrt{(42.4^2 + 21.95^2)} = 47.8 \text{ kN}$$

Bolt capacity M22: $A_{nb} = 0.78 \times 380 = 296.5$; $V_{dsb} = (400/\sqrt{3}) \times 296.5 / 1.25 =$ **54.8 kN** > 47.8 ✓

Bearing: $e = 50$, $p = 80$, $d_0 = 24 \rightarrow k_b = \min(0.694, 0.861, 0.976, 1) = 0.694 \rightarrow V_{dpb} = 125.3$ kN ✓ → **Safe**

(M20 would give 45.3 kN < 47.8 → unsafe.)

Case B – Load perpendicular to plane of bolt group (TENSION + SHEAR)

Bracket bolted to column flange; vertical load P at eccentricity e from the flange face.

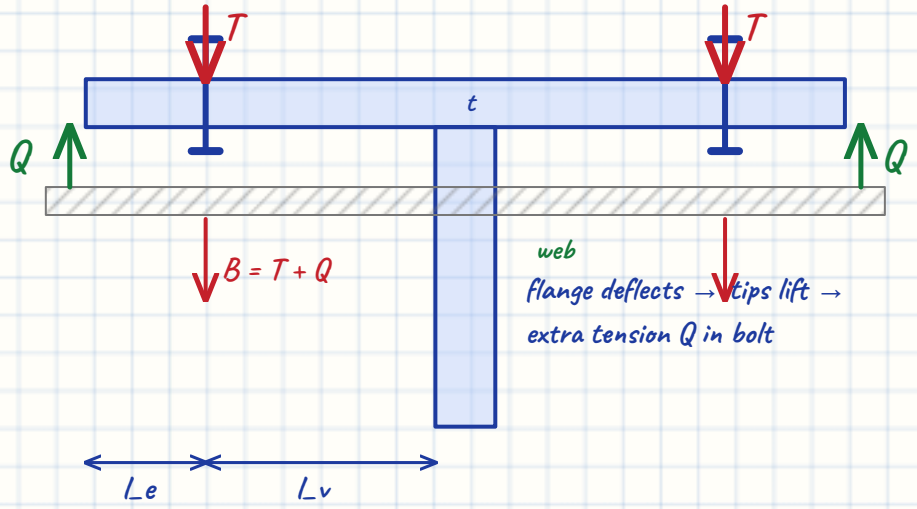
- Direct shear per bolt $V = P/n$.
- Moment $M = P\,e$ rotates the bracket about a **neutral axis**. Taking first the bottom bolts in tension:

$$T_i = M\,y_i / \Sigma\, y^2 \text{; (y measured from the point of rotation/bottom of bracket for simple approx.)}$$

- Check combined: $(V/V_{db})^2 + (T/T_{db})^2 \le 1$.
- If the end-plate is stiff and bolts are not pre-tensioned, the neutral axis is taken at the **bottom flange** (conservative) for a deep bracket; else at centroid of bolt group for short brackets.

Bolt in tension under **prying** → add Q (see next sections).

10. Prying Action



T-stub connection – prying force Q develops at flange tips

When a flexible connecting plate (T-stub, end plate, hanger) is pulled by bolts, the plate **bends**; its tips press on the support and give extra reaction **Q** → bolt force becomes **T + Q**.

IS 800 (Cl. 10.4.7) approach

$$Q = (l_v / 2l_e)\,[T_e - (\beta\,\eta\,f_o\,b_e\,t^4) / (27\,l_e\,l_v^2)]$$

- T_e = external tension per bolt, t = flange thickness, b_e = effective width of flange per bolt
- l_v = distance from bolt centre to toe of fillet weld / root of web; l_e = min(e, 1.1 $t\sqrt{(\beta f_o/f_y)}$)
- β = 2 (non-pretensioned bolts), 1 (pretensioned); η = 1.5; f_o = 0.7 f_y
- Provided: $T_e + Q \le T_{db}$ (bolt tension capacity)

Remedies / Reducing prying

- Thicker flange ($t\uparrow$) – prying $\propto 1/t^3$; bolts closer to web ($l_v\downarrow$)
- Larger edge distance, stiffeners, pre-tensioned (HSFG) bolts

> Prying is **neglected** when the flange is thick enough ($t \ge \approx 1.1\sqrt{(\dots)}$) or bolts are HSFG pre-tensioned & joint is designed as non-flexible.

11. HSFG (High Strength Friction Grip) Bolts

Made of high-strength steel (grades 8.8, 10.9), tightened to a **specified initial tension F_o = 0.7 f_{ub} A_{nb}**. Plates are clamped → load passes by **friction**, bolts are not in bearing.

Advantages

- No slip → **rigid joint**, good for fatigue, vibration, load reversal (bridges, crane girders)
- Bolt not in shear/bearing (pre-tension carries); less number of bolts than black bolts at service
- Quiet, easy to inspect with **torque/turn-of-nut** methods

Design slip resistance (IS 800 Cl. 10.4.3)

$$V_{dsf} = V_{nsf}/\gamma_{mf} ; V_{nsf} = \mu_f n_e K_h F_o$$

- μ_f = coefficient of friction (Table 20): **0.33** clean mill scale, **0.48** sand/shot-blasted, 0.10 red-lead painted...
- n_e = number of effective interfaces (friction surfaces) ; K_h = 1.0 (std. holes), 0.85 (oversize / short slot), 0.7 (long slot)
- γ_{mf} = 1.25 (1.10 if slip is serviceability limit)
- Tightening: turn-of-nut, calibrated wrench, load-indicating washers (DTI).

Tension + shear

$$V_{sf}/V_{dsf} + T_e/T_{ndf} \le 1.0 ; T_{ndf} = 0.9 F_o/\gamma_{mf}$$ (slip-resistance reduced by external tension)

EXAMPLE 4 – HSFG BOLT VALUE

M20 grade 8.8 HSFG, mill scale surfaces, std. holes, 2 friction surfaces:

A_{nb} = 245 mm² ; F_o = 0.7 × 800 × 245 = **137.2 kN**

V_{dsf} = 0.33 × 2 × 1.0 × 137.2 / 1.25 = **72.4 kN** per bolt (slip resistance).

Ultimate (bearing) check – design also for V_{db} at ULS if joint is slip-critical only at service.

12. Welded Connections – Basics

Weld = fusion of parent metal (and filler electrode) by heat – **arc welding**: shielded metal arc (SMAW), submerged arc (SAW – automatic, shop), MIG/TIG.

Advantages vs. bolts

- No holes → full section effective ; lighter (no splice plates), neat, **rigid** joints
- Quiet, can join any shape, less fabrication labour

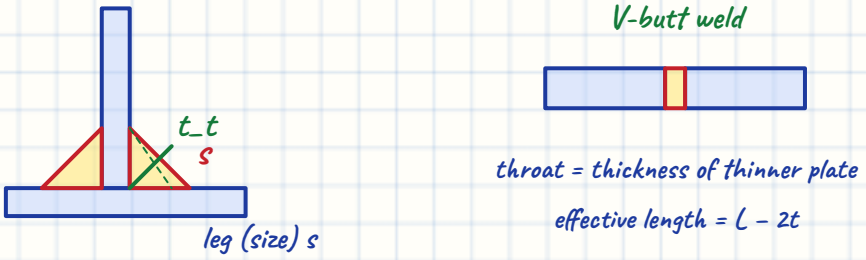
Disadvantages

- Skilled labour & inspection (NDT), **residual stresses, distortion**, fatigue crack, brittle fracture risk, difficult in field

Types of welds

- **Fillet weld** (≈ 80 % of structural welds) – triangular; used in lap & T-joints
- **Butt (groove) weld** – full or partial penetration: square, V, double-V, U, bevel
- Plug / slot weld (rare), spot weld

Joints: lap, butt, T, corner, edge.



throat $t_t = 0.7\ s$ (45° fillet)

Fillet weld – leg size s and effective throat t_t; V-butt weld

Fillet weld rules (IS 800 Cl. 10.5)

- Throat t_t = **K·s**, K = **0.7** (angle between fusion faces 60°–90°) ; 0.6 for 91°–100° ; 0.5 for 101°–106°...
- **Min. size** (by thicker part): ≤10 mm → 3 mm ; 10–20 → 5 mm ; 20–32 → 6 mm ; 32–50 → 8 mm (first run 10 mm)
- **Max. size**: edge of plate → t – 1.5 mm ; for rolled sections beyond.. ≤ thickness of thinner part
- **Min. effective length** ≥ 4 s ; Effective length = overall – 2 s
- Overlap in lap joint ≥ **4 × thickness of thinner plate** (or 40 mm)
- Longitudinal fillet welds on flat bar ends: L ≥ transverse spacing (b) ; end returns ≥ 2s
- Long welds: if $L_j > 150\ t_t$ → reduce $\beta_{lw} = 1.2 - 0.2\ L_j/(150\ t_t)$ (≥ 0.6)

13. Strength of Welds

Fillet weld (shear on throat plane)

$$f_{wd} = f_u / (\sqrt{3} \cdot \gamma_{mw}) \; (f_u = \text{smaller of weld / parent metal})$$
$$P_{dw} = f_{wd} \cdot t_t \cdot L_w \text{ (per weld)} ; \text{ strength per mm} = 0.7\,s\,f_{wd}$$

For Fe 410: shop weld $\gamma_{mw} = 1.25 \rightarrow f_{wd} = 410/(\sqrt{3}\times1.25) =$ **189.4 MPa** ; field weld $\gamma_{mw} = 1.5 \rightarrow$ **157.8 MPa**.

Strength per mm of 6 mm shop fillet = 0.7×6×189.4 = **795 N/mm**.

Butt weld

- Full penetration: strength = strength of parent metal; throat = thickness of thinner part (butt weld is "as strong as plate").
- Design stress in tension/compression = f_y/γ_{m0}, shear = $f_y/(\sqrt{3}\,\gamma_{m0})$ of weaker part.
- Incomplete penetration: throat = ≈ 5/8 × thickness, etc. – eccentricity & stress concentration.
- Effective length = L – 2t (no run-off plates)

EXAMPLE 5 – LAP JOINT WITH FILLET WELDS

Two plates 200 × 12 mm lapped, P_u = **400 kN**, shop welds, Fe 410. Choose s = 8 mm (≥ 5 mm min ; ≤ 12 – 1.5).

t_t = 0.7 × 8 = 5.6 mm ; strength/mm = 5.6 × 189.4 = 1060 N/mm

Required effective length L_w = 400 000/1060 = **377 mm**

Use two longitudinal side welds + end return : each side L_{eff} = 189 mm → overall = 189 + 2s = **205 mm** (~210 mm each side).

Check overlap ≥ 4t = 48 mm ✓ ; L ≥ b = 200 ✓ (210 mm).

14. Eccentric Welded Connections

Case 1 – In-plane eccentricity (TORSION + SHEAR) – weld group treated as a line

- Direct shear stress (per unit length): q_d = P / L_total (parallel to P)
- Torque T = P·e about centroid G of the weld group
- Torsional force per unit length at point at distance r: $q_T = T\,r / I_p$; $I_p = I_{xx} + I_{yy}$ (of weld **lines**)
- Resultant $q_R = \sqrt{[(q_d + q_{T,v})^2 + q_{T,h}^2]}$ at the critical end point.

$$\text{Need: } q_R \le f_{wd}\,t_t \text{ (per mm length)} \rightarrow \text{choose size s.}$$

Case 2 – Out-of-plane eccentricity (BENDING + SHEAR)

Bracket welded to column face by two vertical welds each L = 200 mm:

- Shear stress $q = P/(2\,L\,t_t)$; bending stress $f_b = M / Z_w$ with $Z_w = 2\,(t_t\,L^2/6) = t_t\,L^2/3$

$$f_{res} = \sqrt{(q^2 + f_b^2)} \le f_{wd}$$

EXAMPLE 6 – BRACKET ON COLUMN FACE

$P_u = 100$ kN, $e = 150$ mm, two vertical welds, $L = 200$ mm each, shop welds ($f_{wd} = 189.4$ MPa).

$q = 100\,000 / (2 \times 200 \times t_t) = 250 / t_t$

$M = 100\,000 \times 150 = 15 \times 10^6$ N·mm ; $Z_w = t_t \times 200^2/3 \rightarrow f_b = 15\times10^6 \times 3 / (40000\,t_t) = 1125 / t_t$

$f_{res} = \sqrt{(250^2 + 1125^2)}/t_t = 1152 / t_t \le 189.4 \rightarrow t_t \ge$ **6.08 mm** $\rightarrow s = 6.08/0.7 = 8.7$ mm

Provide fillet weld size 10 mm (round up ; check min size ✓).

Reminder – Always check **parent metal** (plate shear / bending) near the weld, and use $\gamma_{mw} = 1.5$ for site welding.

Quick comparison: Bolted vs Welded

Item	Bolted	Welded
Fabrication	needs holes, plates	direct, fewer parts
Site work	easy	difficult, needs power
Net area	reduced	full
Fatigue	better (HSFG)	poorer
Inspection	visual	NDT required